Exam 1

MATH 2025

Instructions

- Exam length: 75 minutes. This exam must be completed in class and entirely by hand. Show your work where appropriate.
- You may use a non-programmable calculator and the provided formula sheet. No other aids (notes, textbooks, computers, phones) are permitted.
- Read each question carefully. If a question appears ambiguous, state your interpretation clearly and proceed.
- Unless explicitly prompted, you are not required to carry out lengthy calculations to decimal answers; exact values, simplified algebraic expressions, or clearly set-up calculations earn full credit.
- Do **not** write any code from scratch. When code is presented, interpret or analyze it as directed.
- Clearly label each answer. Partial credit may be awarded when reasoning is shown.

The exam is divided into four sections. Suggested time allocations are provided to help you manage the 75-minute limit.

Section A. Multiple Choice (Suggested time: 20 minutes)

For each question, circle the single best answer. Each question is worth 4 points.

1. Which step of the data analysis cycle focuses on transforming raw data into a tidy, analysable form?

    - (A) Pose a question
    - (B) Obtain data
    - (C) Wrangle/clean data

2. When creating scatterplots, which R function from is primarily used to build the plot?

   - (A) `gf_plot()`
   - (B) `gf_point()`
   - (C) `gf_scatter_plot()`

3. Suppose we fit a simple linear regression model to predict monthly apartment rent (in hundreds of dollars) from size (in hundreds of square feet). Which of the following statements about the slope coefficient is **true**?

   - (A) It represents the average change in size for a one-dollar increase in rent.
   - (B) It represents the predicted change in rent for a one-foot increase in size.
   - (C) It is guaranteed to be positive whenever there is a positive correlation.

4. When conducting a randomization (simulation-based) hypothesis test for a regression slope, which option best describes how the null distribution is generated?

   - (A) Resampling residuals with replacement and refitting the model many times under the assumption the null hypothesis is true.
   - (B) Shuffling the response values relative to the predictor and refitting the model many times to represent no relationship.
   - (C) Using a theoretical t distribution with $n-2$ degrees of freedom.

5. Consider the following R output from a model relating park accessibility points (`park_points`) to spending per resident (in hundreds of dollars):

```
#> Coefficients:
#> (Intercept)     35.2
#> spend_per_res   4.8
#> ---
#> Residual standard error: 6.9 on 98 degrees of freedom
```

   What is the predicted `park_points` score for a city that spends \$150 per resident?

   - (A) $35.2 + 4.8 \times 1.5 = 42.4$
   - (B) $35.2 + 4.8 \times 150 = 755.2$
   - (C) $4.8 \times 1.5 = 7.2$
   - (D) Need more information to compute the prediction.

6. Which statement correctly distinguishes a prediction interval from a confidence interval for the mean response at the same predictor value?

   - (A) A prediction interval is narrower because it only concerns a single future case.
   - (B) A prediction interval accounts for both the uncertainty in the regression line and the variability of individual responses.
   - (C) Both intervals have identical widths when the model conditions hold.

- (D) Only a confidence interval requires the equal variance (constant spread) condition.

Section B. Short Answer (Suggested time: 20 minutes)

Show all reasoning. Each question is worth 8 points.

1. **Conditions for Simple Linear Regression**: List and briefly describe the four core conditions for simple linear regression. Provide a concise explanation of how you would check each condition using plots or summaries.

2. **Interpreting Model Output**: A linear model was fit to predict annual cycling miles (`miles`) from the number of bike-share stations in a city (`stations`). The R output below summarizes the fitted model. Interpret the slope, intercept, and R^2 in the context of the problem. Comment on whether the intercept has a meaningful interpretation.

```
#> Call:
#> lm(formula = miles ~ stations, data = bikes)
#>
#> Coefficients:
#> (Intercept)    120.5
#> stations         8.3
#>
#> Residual standard error: 35.6 on 48 degrees of freedom
#> Multiple R-squared:  0.62, Adjusted R-squared:  0.60
```

3. **Residual Patterns**: A residual vs. fitted plot for a commuting time model shows a clear funnel shape, with residual spread increasing as fitted values increase. Which model condition is being violated? Explain how this impacts the reliability of slope inference and suggest one possible remedy.

4. **Confidence Interval Interpretation**: A simulation-based procedure for the slope produced a 95% confidence interval of $(1.2, 2.8)$ minutes per mile when predicting commute time from travel distance. State what this interval means in context, including the interpretation of the confidence level.

Section C. True/False with Justification (Suggested time: 15 minutes)

For each statement, circle True or False and justify your answer in 1–2 sentences. Each question is worth 5 points.

1. **True / False**: If the correlation between study hours and exam score is near zero, then fitting a simple linear regression is pointless because the slope must also be exactly zero.
2. **True / False**: When creating a scatterplot in `ggplot`, swapping the roles of the explanatory and response variables on the axes changes the underlying correlation between them.
3. **True / False**: In a randomization test for the slope, a p-value of 0.03 means there is only a 3% chance the alternative hypothesis is true.
4. **True / False**: A t-based confidence interval for the slope relies on the sampling distribution of the slope being approximately normal, which is more plausible when the sample size is large or the residuals are nearly normal.
5. **True / False**: Prediction intervals widen as you move away from the mean of the predictor values even when the constant spread condition is satisfied.

Section D. Evaluate a Response (Suggested time: 20 minutes)

Each item below presents a student's written response. Evaluate the response by (i) stating whether the student's conclusion is *correct*, *partially correct*, or *incorrect*, and (ii) explaining your reasoning. Each question is worth 10 points.

1. **Checking Linearity**: Student A writes, "The scatterplot of city population versus public library visits looks mostly straight except for one point far above the line, so the linearity condition is satisfied." Evaluate this claim, commenting on the influence of the outlier and whether additional investigation is needed.
2. **Hypothesis Test Conclusion**: Student B used a randomization test for the slope in a model relating greenhouse gas emissions to average household energy bills and obtained a p-value of 0.18. They conclude, "Because the p-value is larger than 0.05, there is strong evidence of a positive relationship between the variables." Assess the correctness of this conclusion.

3. **Prediction vs. Confidence Interval**: Student C reports, "A 95% prediction interval of (12, 28) minutes for the next commute means that 95% of all commuters have commute times between 12 and 28 minutes." Evaluate this statement.

Point Summary

Section	Points
A	24
B	32
C	25
D	30
Total	**111**

(Grading may rescale totals to a 100-point scale.)

Solutions

Section A. Multiple Choice

1. **(C)** The wrangle/clean step transforms raw data into a tidy structure that supports analysis.
2. **(B)** `ggplot()` initializes the plotting system, after which aesthetics and geoms are added.
3. **(B)** The slope estimates the average change in rent for a one-unit increase in size.
4. **(B)** Shuffling the response relative to the predictor simulates a world with no association between them.
5. **(A)** Spending $150 corresponds to 1.5 hundreds, so the prediction is $35.2+4.8(1.5) = 42.4$ park points.
6. **(B)** Prediction intervals include uncertainty from both the fitted mean and individual variability, making them wider.

Section B. Short Answer

1. **Regression conditions**: (i) *Linearity* — the relationship between predictor and response is approximately linear; check with a scatterplot or residuals vs. fitted plot. (ii) *Independence* — observations do not influence each other; check study design or order of data collection. (iii) *Constant spread* — residual variance is roughly equal across fitted values; assess with a residuals vs. fitted plot for uniform scatter. (iv) *Normal residuals* — residual distribution is roughly symmetric without extreme tails; assess with a histogram or normal Q-Q plot.
2. **Interpretation**: The slope of 8.3 indicates that, on average, each additional bike-share station is associated with about 8.3 more cycling miles per year. The intercept of 120.5 predicts miles when a city has zero stations; if a zero-station city is outside the data range, the intercept may not be meaningful. The R^2 of 0.62 means about 62% of the variability in annual cycling miles is explained by the number of stations in this model.
3. **Condition violation**: The funnel shape signals a violation of the constant spread (homoscedasticity) condition. Inference on the slope may be unreliable because standard errors are misestimated. A remedy is to transform the response or predictor (e.g., log transformation) or consider weighted least squares to stabilize variance.
4. **Confidence interval meaning**: We are 95% confident that the true increase in commute time per mile of travel lies between 1.2 and 2.8 minutes. Over many similar studies using the same procedure, about 95% of the computed intervals would capture the true slope.

Section C. True/False with Justification

1. **False.** A near-zero correlation suggests a weak linear association but the sample slope need not be exactly zero; sampling variability can produce small nonzero slopes.
2. **False.** Switching axes changes which variable is treated as predictor or response but does not alter the underlying correlation value.
3. **False.** A p-value of 0.03 measures the probability of obtaining a result as or more extreme under the null, not the probability the alternative is true.
4. **True.** t-based intervals rely on the sampling distribution of the slope being approximately normal, which becomes more plausible with large samples or nearly normal residuals.
5. **True.** Prediction intervals account for individual variability and tend to widen for predictor values farther from the sample mean.

Section D. Evaluate a Response

1. **Partially correct.** The scatterplot may look linear, but an influential outlier can distort the perceived trend. The student should investigate the outlier (e.g., by refitting without

it or checking for data issues) before concluding linearity holds.

2. **Incorrect.** A p-value of 0.18 indicates the observed slope is plausible under the null; it does not provide strong evidence for a positive relationship. The student should conclude there is insufficient evidence to claim a positive association.
3. **Incorrect.** A 95% prediction interval applies to a single future commute; it does not mean 95% of all commute times fall within that range. It reflects uncertainty about the next observation.